

QA REPORT

Intern Information

Role: QA Tester Intern (AI/ML Domain)

Day: 5

Date: 17-June-2026

Activities Completed

1. Comprehensive Test Planning

A detailed test plan was prepared covering:

- Functional Testing
- API Testing
- Security Testing
- Performance Testing
- Regression Testing

Deliverables

- Scope Definition
- Test Strategy
- Entry & Exit Criteria
- Risk Assessment
- Test Metrics
- Roles & Responsibilities

Status: Completed

2. Automated Test Case Execution

Module Tested

Login Authentication Module

Coverage

- Valid Login
- Invalid Login
- Empty Fields

- Password Validation
- Session Management
- SQL Injection
- XSS Protection
- Remember Me Feature
- Account Lock Mechanism

Results

Metric	Count
Total Test Cases	20
Executed	20
Passed	18
Failed	2
Pass Rate	90%

Initial Defects Found

1. Long Username Validation
2. Session Timeout
3. Account Lock Mechanism

Status: Defects reported and fixed.

3. API Testing & Validation

Tool Used

Postman

APIs Validated

- GET
- POST
- PUT
- DELETE
- Login API
- Registration API

Validation Areas

- Status Codes
- Response Body
- Response Time
- JSON Structure
- Authentication
- Error Handling

- CRUD Operations

Results

Metric	Count
APIs Tested	8
Passed	8
Failed	0
Pass Rate	100%

Minor Defects Identified

- Inconsistent error message format
- Missing validation for long username input

Status: Open for enhancement.

4. Regression Testing

Objective

Verify that defect fixes do not impact existing functionality.

Modules Retested

- Login
- Logout
- Forgot Password
- Input Validation
- Session Management
- Security Controls

Results

Metric	Count
Regression Tests	20
Passed	20
Failed	0
Defects Reopened	0

Verified Fixes

- Username Length Validation
- Session Timeout
- Account Lock Mechanism

Status: Successfully Closed

5. Usability Testing

Areas Reviewed

- Navigation
- Accessibility
- Form Design
- Error Messages
- Mobile Responsiveness

Findings

- Low sidebar text contrast
- Search icon visibility concerns
- Small mobile action buttons
- Multi-click navigation paths

Status: Improvement recommendations provided.

6. Performance & Load Testing

Performance Results

Scenario	Response Time
Login	1.4 sec
Dashboard Load	3.8 sec
Search	2.5 sec
Report Generation	6.3 sec

Load Results

Users	Result
50	Stable
100	Stable
500	Performance degradation observed

Status: Acceptable for moderate traffic.

7. Document Processing Workflow Validation

Workflow Validated

Upload → Validation → OCR → Verification → Storage → Retrieval → Download

Results

Area	Status
Upload Validation	Pass
OCR Extraction	Pass
Data Verification	Pass
Storage Validation	Pass
Retrieval Validation	Pass

OCR Accuracy

98%

8. Authentication & Authorization Testing

Validated Areas

- Login
- Logout
- Password Reset
- Session Management
- Role-Based Access Control
- Unauthorized Access Prevention

Results

All authentication and authorization test scenarios passed successfully.

Status: Pass

Overall Test Metrics

Category	Executed	Passed	Failed
Automation Testing	20	18	2
API Testing	8	8	0
Regression Testing	20	20	0
Authentication Testing	10	10	0

Authorization Testing	10	10	0
Document Workflow Validation	15	15	0

Total

Metric	Value
Total Tests Executed	83
Passed	81
Failed	2
Overall Pass Rate	97.6%

Defect Summary

Severity Count

Critical 0
High 0
Medium 2
Low 4

Current Status

- Critical Defects: None
 - High Severity Defects: None
 - Regression Defects: Closed
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Recommendations

Functional Quality

- Increase automated test coverage to additional modules.
- Expand boundary value and negative testing scenarios.

API Quality

- Standardize API error response structure.
- Add validation for edge-case inputs.
- Introduce automated API regression suites.

Security

- Continue SQL Injection and XSS testing.
- Conduct periodic vulnerability assessments.

- Implement security testing in CI/CD pipelines.

Performance

- Optimize database queries.
- Introduce caching mechanisms.
- Monitor application performance under peak traffic.

Usability

- Improve accessibility compliance.
- Increase mobile touch target sizes.
- Simplify navigation workflows.

Document Processing

- Improve OCR accuracy for special characters.
- Optimize processing of large documents.
- Enhance duplicate detection capabilities.

Final Conclusion

Day 5 QA activities were completed successfully. Comprehensive test planning, automation testing, API validation, regression testing, usability testing, performance testing, document workflow validation, and security testing confirmed that the application is stable and functionally reliable.

The overall test pass rate achieved was 97.6%, with no critical or high-severity defects remaining open. Previously reported issues were successfully fixed and validated through regression testing.